JAIN Knowledge Campus, Jayanagar 9th Block, Bengaluru

DEPARTMENT OF COMPUTER SCIENCE AND IT

MCA PROGRAMME

ACADEMIC YEAR: 2026-2027

EXP: 02 - AZURE SYNAPSE ANALYTICS WORKSPACE AND SQL POOLS

NAME: ABHISHEK KUMAR USN: 25MCAR0141

SEM: III SPEC: SCT

SUBJECT: CLOUD ANALYTICS-LAB

CODE: 25MCASC302L

FACULTY NAME: Dr. PRABHAKARAN ABRAHAM, ASSOCIATE PROFESSOR

1 Aim

To create an Azure Synapse Analytics workspace, explore its SQL pools, and verify the pool status and SQL request monitoring for the workspace.

2 Objective and Theory

The objectives are to provision the Synapse workspace in the same resource group as the Data Factory and storage used in Experiment 01, to open Synapse Studio, to check the status of the built-in SQL pool, and to monitor SQL requests submitted to the workspace. Azure Synapse Analytics provides two kinds of SQL pool: a dedicated SQL pool, which is provisioned compute charged by the hour, and a built-in serverless SQL pool, which is always available in a workspace and charged per query on the data processed [1]. The work in this experiment used the built-in serverless SQL pool.

A dedicated SQL pool was not created in this work, and no dedicated pool exists in this workspace. That is stated here and in Section 5, and no dedicated pool creation, dedicated pool screenshot, or dedicated pool result is claimed in this report. The reason was cost: a dedicated pool is hourly billable, while the serverless pool used here carries no hourly pool charge. The same fact was verified again in Experiment 08, where the workspace was recreated and no dedicated SQL pool and no Spark pool were created.

3 Architecture Flow

Data and control flow from the resource group to the Synapse workspace, to the built-in serverless SQL pool, to SQL request monitoring, as shown in Figure 1.1. Table 1.1 lists the resources used.

Table 1.1 Resources used in Experiment 02

Component	Resource name	Role
Resource Group	25MCAR0141	Logical container for all experiment resources.
Synapse Workspace	calab-synapse-141	Synapse workspace, Central India, storage calablake141, container rawdata.
SQL Pool	Built-in serverless SQL pool	Built-in pool, shown Running online in Screenshot 4. No dedicated pool created.
Storage	calablake141	ADLS Gen2 storage, primary storage for the workspace.
Data Factory	calab-adf-141	Data Factory from Experiment 01, in the same resource group.

Source: Azure Portal and Synapse Studio screenshots in Section 6, and Experiment 08 workspace verification, 2026-11-01.

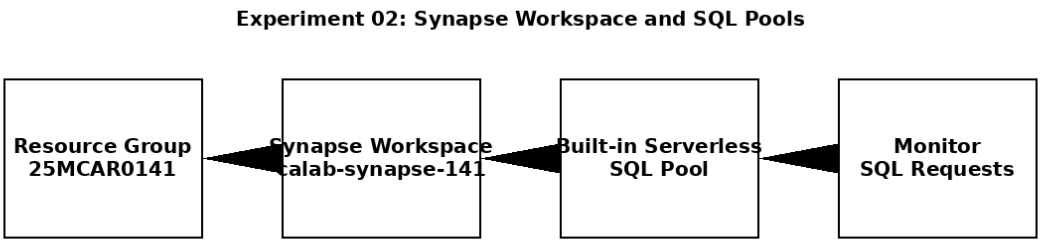


Figure 1.1 Experiment 02 flow

Resource group to Synapse workspace to built-in serverless SQL pool to SQL request monitoring.

4 Procedure

Step 1 (Resource provisioning). Create Resource Group 25MCAR0141 in the Azure Portal and deploy Data Factory calab-adf-141, Storage Account calablake141, and Synapse Workspace calab-synapse-141, as in Experiment 01. The resource groups list and the resource group overview are shown in Screenshots 1 and 2.

Step 2 (Open Synapse Studio). Open Synapse Studio for workspace calab-synapse-141. The workspace landing page is shown in Screenshot 3.

Step 3 (Check the SQL pool). In Synapse Studio, check the built-in SQL pool status. The management view shows the Built-in Serverless SQL pool Running online, as shown in Screenshot 4. No dedicated SQL pool was created, so no dedicated pool status is shown.

Step 4 (Monitor SQL requests). Open the SQL request monitoring view. It shows completed SQL requests with submit times and processing duration, and 10.00 MB data processed, as shown in Screenshot 6.

Step 5 (Verify resources). Verify the deployed resources in the resource group overview, as shown in Screenshot 5.

5 Proof of Work

The GitHub repository for this experiment is https://github.com/25MCAR0141/Cloud-Analytics-Azure/tree/main/Lab/Exp_02 . The repository contains the README, architecture diagram, and the Synapse screenshots copied from Experiment 01, where this work was first recorded. The screenshots are the original Experiment 01 files, Step_01, Step_02, Step_10, Step_14, Step_15, and Step_16, kept under their original names so the source can be traced. Table 1.2 records the workspace and pool.

Table 1.2 Synapse workspace and SQL pool record, Experiment 02

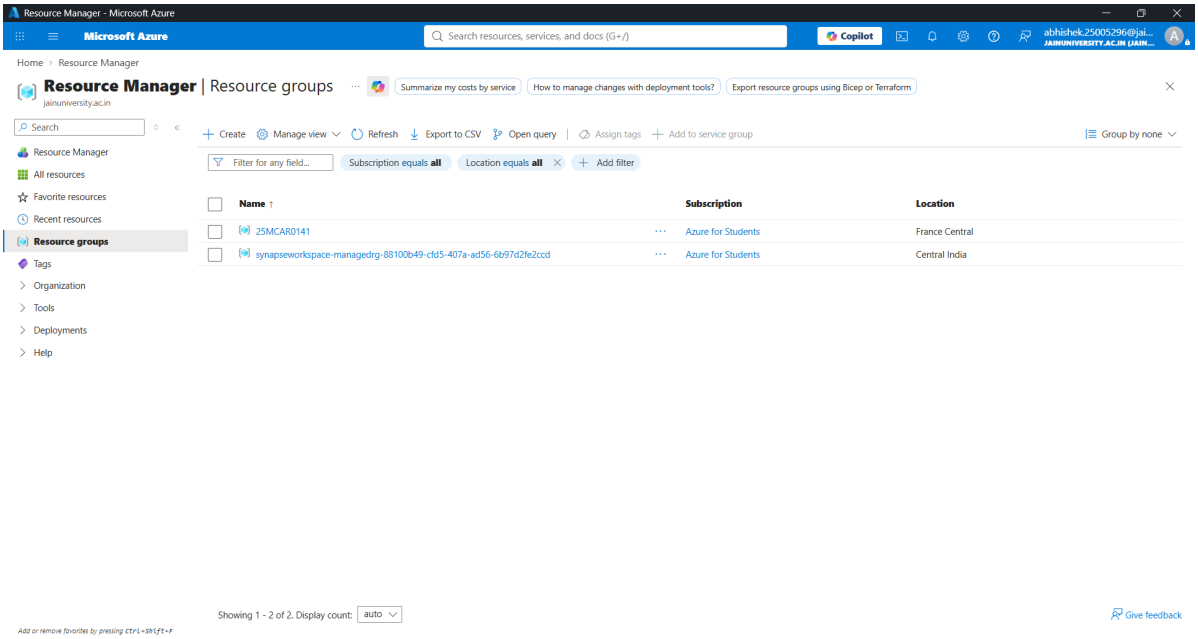
Item	Value
Synapse workspace	calab-synapse-141, Central India, storage calablake141, container rawdata
Serverless SQL endpoint	calab-synapse-141-ondemand.sql.azuresynapse.net
Built-in SQL pool	Built-in Serverless SQL pool, Running online, Screenshot 4
Dedicated SQL pool	Not created, no dedicated pool in the workspace
Spark pool	Not created
SQL monitoring	Completed SQL requests shown, 10.00 MB data processed, Screenshot 6

Source: Synapse Studio screenshots in Section 6, and Experiment 08 workspace record, 2026-11-01.

Result. The Synapse workspace was created and verified, the built-in serverless SQL pool was shown Running online, and SQL requests were monitored with 10.00 MB data processed. A dedicated SQL pool was not created, so this experiment does not demonstrate dedicated pool provisioning.

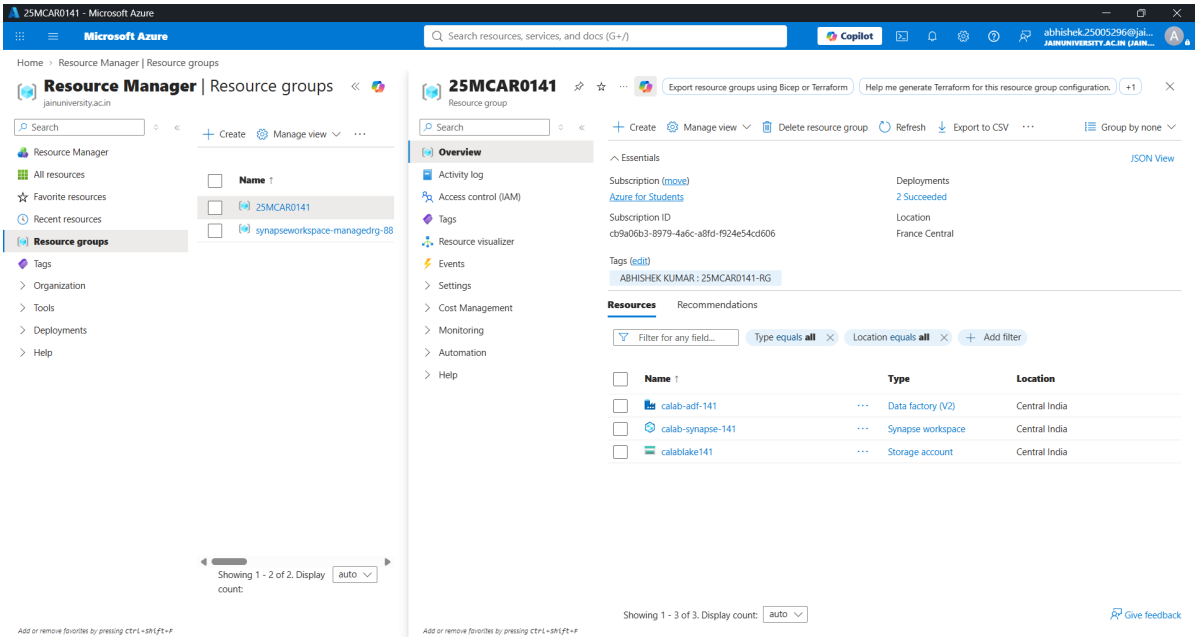
6 Screenshots

The screenshots in this section are the original Experiment 01 screenshots, copied without change, showing the Synapse workspace and SQL pool work that was first recorded under Experiment 01. They are real Azure Portal and Synapse Studio page captures, 1920 by 1020 pixels, embedded at landscape aspect, 6.2 by 3.29 inches. The full Experiment 01 report remains at https://github.com/25MCAR0141/Cloud-Analytics-Azure/tree/main/Lab/Exp_01 .



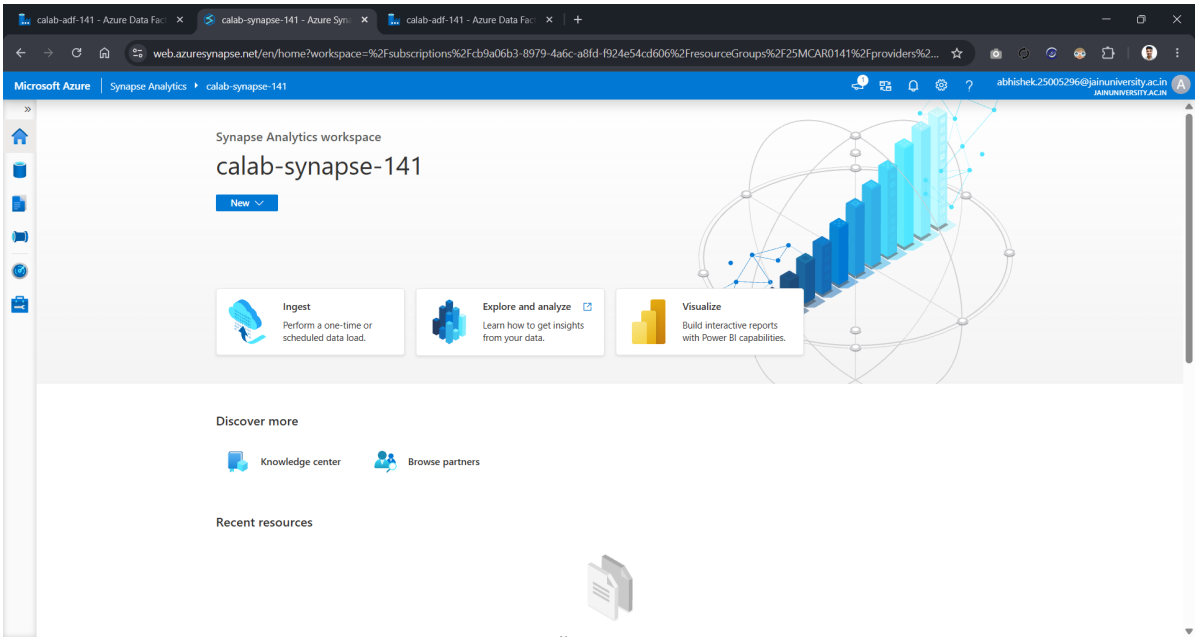
Screenshot 1 Azure Resource Groups list

Resource groups view showing primary resource group 25MCAR0141 and the managed Synapse resource group.



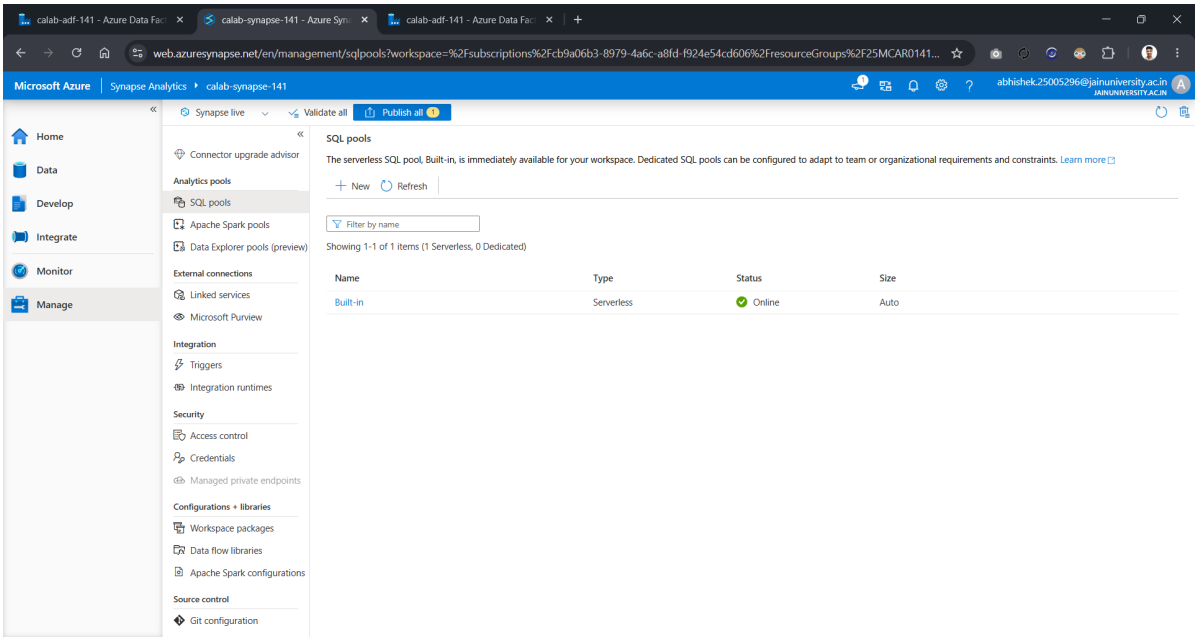
Screenshot 2 Resource Group 25MCAR0141 overview

Overview of deployed resources: Data Factory calab-adf-141, Synapse Workspace calab-synapse-141, and Storage Account calablake141.



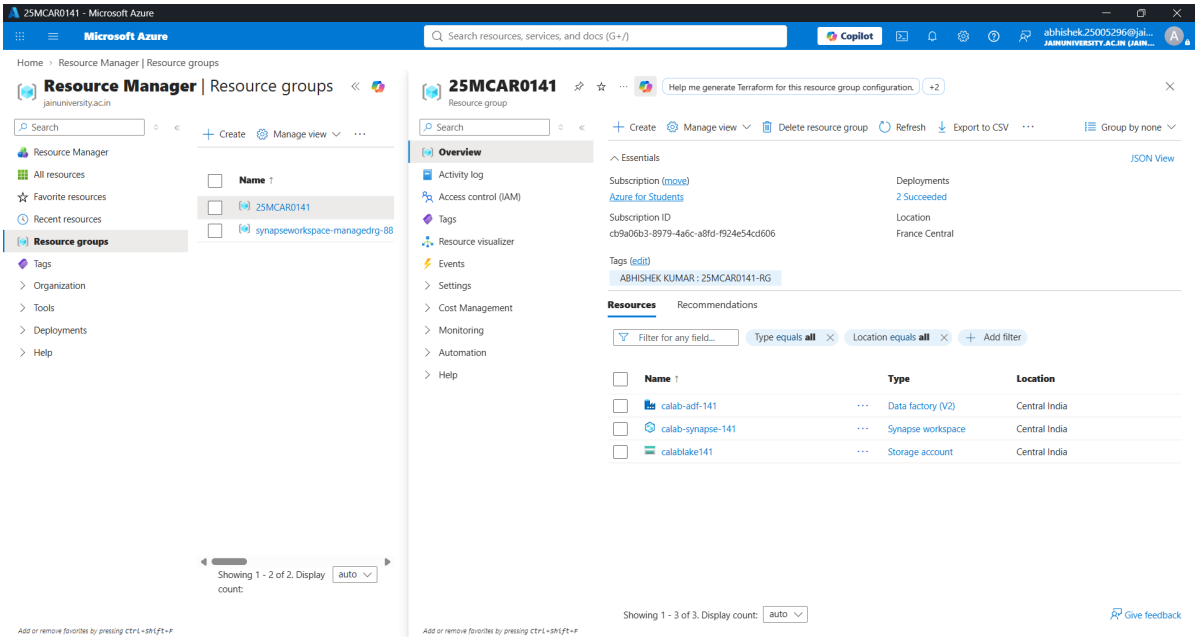
Screenshot 3 Synapse Workspace landing page

Synapse Analytics Studio interface used for serverless SQL data exploration and querying.



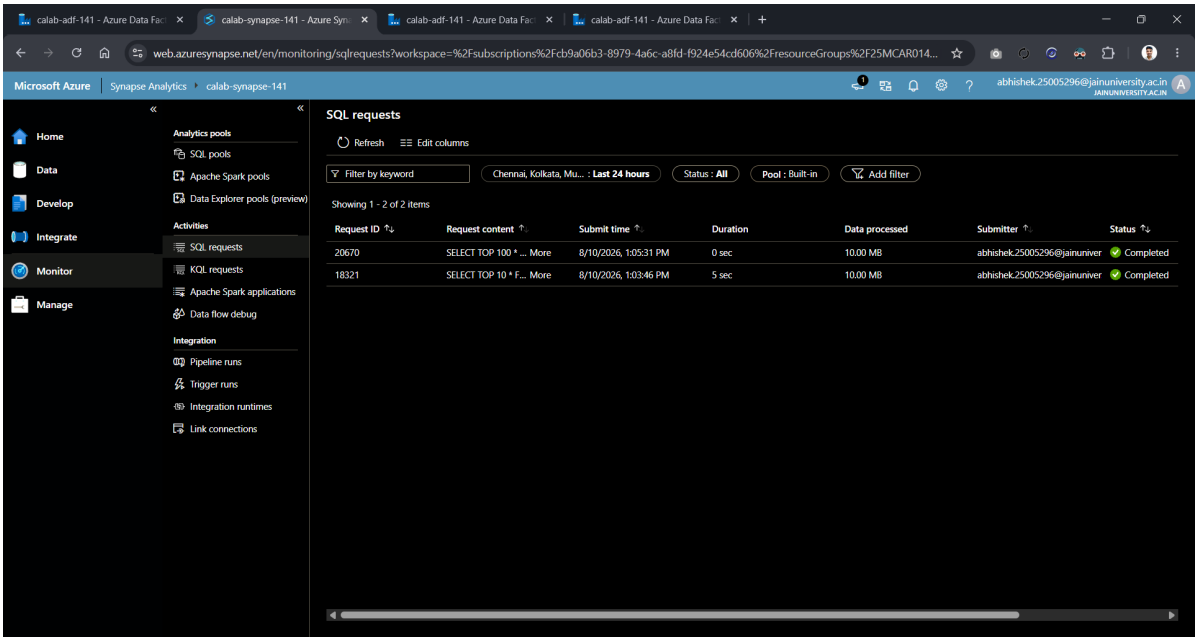
Screenshot 4 Built-in SQL pool status

Management view showing Built-in Serverless SQL pool Running online. No dedicated pool is shown, because none was created.



Screenshot 5 Resource Group final verification

Final validation of active Azure resources deployed under subscription Azure for Students.



Screenshot 6 Synapse SQL request monitoring log

Monitoring view showing completed SQL requests, submit times, processing duration, and 10.00 MB data processed.

Cost note. No hourly pool charge applies in this experiment, because no dedicated SQL pool and no Spark pool were created. The built-in serverless SQL pool is charged per query on data processed, and the monitoring view shows 10.00 MB data processed.

References

[1] Microsoft, "Azure Synapse Analytics documentation," Microsoft Learn. Available: https://learn.microsoft.com/en-us/azure/synapse-analytics/

[2] Microsoft, "Azure Data Lake Storage Gen2 documentation," Microsoft Learn. Available: https://learn.microsoft.com/en-us/azure/storage/blobs/data-lake-storage-introduction